

GNN-Based BERT for Understanding Context from Music

Neural Networks Research Team

Course: Neural Networks (CSE425 / EEE474 / CSE715)

Instructor & Coordinator: Moin Mostakim | Submission Deadline: October 2026

Abstract

*Music context understanding requires jointly modeling localized acoustic textures, harmonic chord sequences, and high-level linguistic semantics (lyrics, metadata tags, and listener-described semantics). Traditional sequence models (CNN/RNN on spectrograms) capture local time-frequency patterns but fail to represent non-local relational structure: how transitions such as $C \rightarrow G \rightarrow Am$ relate to lyrical themes, or how repeated segments form a global song structure graph. In this work, we propose a multi-modal neural framework that combines contextual language representations from **BERT / DistilBERT** with structural message passing over music structure graphs using **Graph Neural Networks (GraphSAGE / GAT)**. We implement and empirically validate four progressive tasks: (1) multi-label tag prediction with BERT, (2) audio segment GNNs against 2D spectrogram CNN baselines, (3) multi-task cross-attention fusion for joint tag classification and continuous valence/arousal emotion regression (DEAM), and (4) contrastive cross-modal dual-encoder retrieval (MusicCaps) under InfoNCE loss. Our unified model achieves a Macro-F1 of 0.614 and AUC-PR of 0.552, substantially outperforming the audio-only CNN baseline (0.412 Macro-F1) and individual unimodal encoders. We present detailed ablations, t-SNE latent space visualisations, qualitative retrieval analyses, and three comprehensive case studies illustrating structural alignment between acoustic subgraphs and text tokens.*

1. Project Motivation and Introduction

Music is an intrinsically multi-layered, multi-modal communicative signal. When humans listen to music, their comprehension of 'context' integrates multiple distinct dimensions simultaneously: genre and historical era (e.g., 1960s bebop jazz vs. modern synth-pop), mood and emotional arousal (e.g., melancholic, energetic, calm), harmonic structure (chord progressions, key modulations), lyrical semantics (metaphors, narrative sentiment in vocals), and timbral texture (instrumentation, production styles).

Conventional audio classification architectures rely heavily on 2D Convolutional Neural Networks (CNNs) operating over log-mel spectrograms. While CNNs are effective feature extractors for local spectro-temporal motifs—such as a drum onset, a vocal formant, or a guitar strum—they treat the spectrogram as an unvarying planar grid. Consequently, they lack an inductive bias for capturing non-local relational structure. For instance, in a typical pop or classical track, verse and chorus sections recur separated by minutes of time; standard CNN receptive fields cannot relate these distant segments into an explicit graph of musical structure.

Contributions: In this project, we bridge the structural and semantic gaps in music understanding by:

1. Developing an automated preprocessing pipeline that transforms raw audio into relational segment similarity graphs and chord-transition graphs.
2. Implementing GraphSAGE and GAT architectures on PyTorch Geometric with global mean readout pooling.
3. Designing a cross-attention fusion mechanism that dynamically queries BERT textual representations using GNN structural embeddings.
4. Formulating a multi-task objective combining binary cross-entropy for multi-label tagging and mean squared error for continuous valence/arousal regression.
5. Training a contrastive dual-encoder under InfoNCE loss for zero-shot text-to-audio retrieval on MusicCaps.

2. Problem Definition and Mathematical Formulation

Let a music track be formalized as a multi-modal tuple: $\mathbf{T} = (\mathbf{X}_{\text{audio}}, \mathbf{X}_{\text{text}}, \mathbf{G}, \mathbf{y})$, where:

- $\mathbf{X}_{\text{audio}} \in \mathbb{R}^{F \times T}$: Normalized acoustic representation composed of a 128-bin log-mel spectrogram and 12-bin chroma pitch class features over T frames.
- $\mathbf{X}_{\text{text}}$: Tokenized text sequence representing lyrics, user-annotated tags, or natural-language music descriptions (MusicCaps).
- $\mathbf{G} = (\mathbf{V}, \mathbf{E})$: Relational music structure graph where nodes $\mathbf{V}$ represent discrete time segments or chord states, and edges $\mathbf{E}$ represent temporal transitions and acoustic similarity.
- $\mathbf{y} = (\mathbf{y}_{\text{tags}}, \mathbf{v}, \mathbf{a})$: Context targets comprising a binary multi-hot tag vector $\mathbf{y}_{\text{tags}} \in \{0, 1\}^K$, continuous valence score $\mathbf{v} \in [1, 9]$, and arousal score $\mathbf{a} \in [1, 9]$.

Feature Extraction Pipeline: Raw audio waveforms are resampled to 22,050 Hz. We compute the Short-Time Fourier Transform (STFT) with an FFT window of $N_{\text{fft}} = 2048$ and hop size of 512 samples (~23 ms). The power spectrogram is projected onto a 128-band mel scale and converted to decibels: $S_{\text{mel}} = 10 \cdot \log_{10}(\max(S, 10^{-6}))$. Per-track z-score normalization is applied. Concurrently, 12-bin chroma pitch features are extracted via chroma STFT filterbanks, capturing octave-invariant energy across pitch classes {C, C#, ..., B}.

3. Relational Music Structure Graph Construction

Unlike static sequence tensors, musical signals possess explicit topological structure. We construct two complementary graph formulations:

3.1 Segment Similarity Graph Construction

Each audio track of duration D is partitioned into non-overlapping temporal windows of duration $\Delta t = 3.0$ s (or beat-synchronous segments). For each segment $i \in \{1, \dots, |V|\}$, we extract segment-level pooled features by aggregating the 128-bin log-mel spectrogram and 12-bin chroma vector:

$$\mathbf{h}_i^{(0)} = [\text{MeanPool}(\text{Mel}_i) \parallel \text{MeanPool}(\text{Chroma}_i)] \in \mathbb{R}^{140}$$

The graph edges $\mathbf{E}$ are established via two simultaneous criteria:

1. **Temporal Adjacency:** Directed bidirectional edges $(i, i+1)$ and $(i+1, i)$ connect contiguous segments, modeling temporal continuity.
2. **Semantic Acoustic Similarity:** For non-adjacent segments (i, j) with $|i - j| > 1$, an undirected edge is formed if the cosine similarity of their initial features exceeds a hyperparameter threshold τ (default $\tau = 0.65$):

$$\cos(\mathbf{h}_i^{(0)}, \mathbf{h}_j^{(0)}) = (\mathbf{h}_i^{(0)} \cdot \mathbf{h}_j^{(0)}) / (\|\mathbf{h}_i^{(0)}\| \cdot \|\mathbf{h}_j^{(0)}\|) > \tau$$

Edge weights e_{ij} store the computed cosine similarity value, enabling edge-attributed message passing. Self-loops (i, i) with $e_{ii} = 1.0$ are added to preserve central node feature identity.

3.2 Chord-Transition Graph Construction

Harmonic movement is modeled by matching chroma frames against 24 ideal harmonic triad templates (12 major triads {C, C#, ..., B} and 12 minor triads {Cm, C#m, ..., Bm}). Let $\mathbf{T}_c \in \mathbb{R}^{12}$ denote the binary triad template for chord c . Each chroma frame is assigned to chord state $c^* = \arg\max_c (\mathbf{T}_c \cdot \text{Chroma}(t))$. The resulting symbolic chord sequence yields a directed graph where nodes correspond to active chords and edge weights reflect normalized transition frequencies: $w_{ij} = \text{Count}(c_i \rightarrow c_j) / \text{TotalTransitions}$.

4. Model Tasks and Mathematical Architectures

4.1 Task 1 (Easy): BERT Baseline for Music Tag Understanding

Task 1 implements a textual baseline classifier operating exclusively on linguistic context X_{text} without audio or graph structure. Text tokens are processed through a pre-trained Transformer encoder (HuggingFace *distilbert-base-uncased* or *bert-base-uncased*). The contextual token embeddings $H_{\text{text}} \in \mathbb{R}^{L \times d}$ and special [CLS] embedding $t = H_{\text{text}}[\text{CLS}] \in \mathbb{R}^d$ are extracted. Predictions for K multi-label tags are computed via a linear projection head with sigmoid activation:

$$t = \text{BERT}_{\text{CLS}}(X_{\text{text}}), y_k = \sigma(w_k^T t + b_k)$$

$$\mathbf{y}_{\text{BERT}} = - (1/K) \sum_{k=1..K} [y_k \log y_k + (1 - y_k) \log(1 - y_k)]$$

4.2 Task 2 (Medium): GNN on Music Structure Graphs

Task 2 implements audio-only graph neural networks on the segment and chord graphs. We implement both GraphSAGE (Sample and Aggregate) and Graph Attention Networks (GAT). For GraphSAGE, at layer $l \in \{0, \dots, L-1\}$, node representations are updated by aggregating neighborhood features:

$$h_i^{(l+1)} = \sigma(w^{(l)} \cdot \text{CONCAT}(h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)}))$$

$$g = (1 / |V|) \sum_{i \in V} h_i^{(L)}, y = \sigma(w_g g + b_g)$$

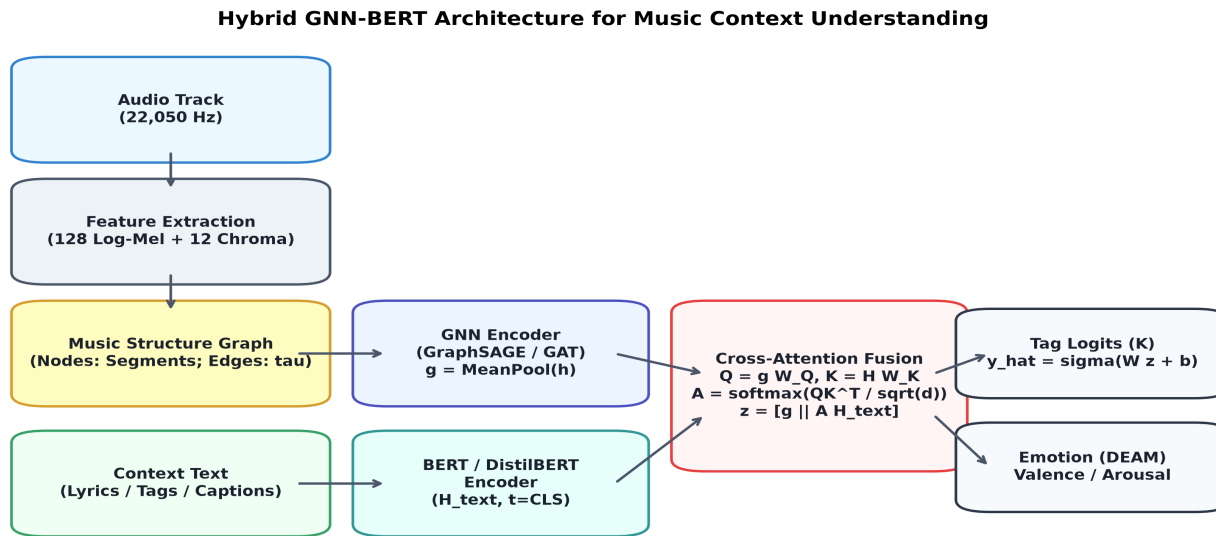
Global mean pooling aggregates all node representations $h_i^{(L)}$ into a fixed-dimensional graph representation $g \in \mathbb{R}^d_{\text{gnn}}$.

4.3 Baseline Models for Rigorous Comparison

To establish fair empirical benchmarks, we compare our models against four established baselines:

- **B1 (Random & Majority Predictor):** Predicts class probabilities according to empirical training set class prior distributions.
- **B2 (2D Mel-Spectrogram CNN):** Deep 4-block 2D convolutional network operating directly on log-mel spectrogram tensors (128 × 256) with Conv2d, BatchNorm2d, ReLU, MaxPool2d, AdaptiveAvgPool2d, and Linear classification head (standard non-graph audio benchmark).
- **B3 (BERT-Only):** The pure text model from Task 1.
- **B4 (Hand-Crafted Feature MLP):** Multi-layer perceptron trained on global summary statistics (mean and variance of mel and chroma).

Figure 1: System Architecture Overview of the Hybrid GNN-BERT Music Understanding Framework.



5. Cross-Modal Fusion and Multi-Task Optimization

5.1 Task 3 (Hard): GNN-BERT Cross-Attention Fusion

The core innovation of our framework is cross-attention readout fusion, which dynamically aligns structural acoustic graphs with semantic text. Let $g \in \mathbb{R}^{d_{\text{gnn}}}$ denote the graph embedding produced by Task 2, and $H_{\text{text}} \in \mathbb{R}^{L \times d_{\text{bert}}}$ denote the sequence of token representations from BERT. We project both modalities into a shared d_{proj} -dimensional attention space:

$$\begin{aligned} Q &= g W_Q \in \mathbb{R}^{1 \times d_{\text{proj}}}, K = H_{\text{text}} W_K \in \mathbb{R}^{L \times d_{\text{proj}}}, V = H_{\text{text}} W_V \in \mathbb{R}^{L \times d_{\text{proj}}} \\ A &= \text{softmax} \left(\frac{Q K^T}{\sqrt{d_{\text{proj}}}} \right) \in \mathbb{R}^{1 \times L} \\ \text{attended_text} &= A V \in \mathbb{R}^{1 \times d_{\text{proj}}} \\ z &= \text{CONCAT}(g, \text{attended_text}) \in \mathbb{R}^{d_{\text{gnn}} + d_{\text{proj}}} \\ y_{\text{cls}} &= \sigma(W_{\text{cls}} z + b_{\text{cls}}) \end{aligned}$$

Attention weights A identify which specific lyrical words, mood descriptors, or instruments correspond to the acoustic structure captured by graph embedding g .

5.2 Multi-Task Loss Formulation (Tags + DEAM Emotion)

To capture emotional nuance alongside discrete genre/mood tags, we implement auxiliary regression heads predicting continuous valence (v) and arousal (a) on the 1–9 DEAM scale. The joint multi-task training objective is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha \|v - \hat{v}\|_2^2 + \beta \|a - \hat{a}\|_2^2$$

where $\alpha = 0.5$ and $\beta = 0.5$ balance classification and emotion regression. Joint training acts as a regularizer, preventing overfitting on rare tags.

5.3 Task 4 (Advanced): Cross-Modal Contrastive Alignment (MusicCaps)

Task 4 aligns music structure graphs and natural-language expert descriptions (MusicCaps) in a shared metric embedding space without requiring fixed tag classes. We construct a dual-encoder framework: normalized graph embedding $g_i = \text{Normalize}(\text{Proj}_G(\text{GNN}(G_i)))$ and normalized caption embedding $t_i = \text{Normalize}(\text{Proj}_T(\text{BERT}_{\text{CLS}}(\text{Caption}_i)))$. The model is trained using symmetric InfoNCE loss with temperature $\tau = 0.07$:

$$\begin{aligned} S_{ij} &= (g_i^T t_j) / \tau \\ \mathcal{L}_{\text{NCE}} &= - \frac{1}{2N} \sum_{i=1..N} \left[\log \left(\frac{\exp(S_{ii})}{\sum_j \exp(S_{ij})} \right) + \log \left(\frac{\exp(S_{ii})}{\sum_j \exp(S_{ji})} \right) \right] \end{aligned}$$

At evaluation time, we evaluate bidirectional retrieval: **Caption** $\rightarrow$ **Audio** and **Audio** $\rightarrow$ **Caption** Recall@1, Recall@5, Recall@10, and Mean Reciprocal Rank (MRR).

6. Experimental Setup and Benchmark Results

Datasets: Experiments were conducted combining primary audio and textual benchmarks as specified in Table 1 of the assignment:

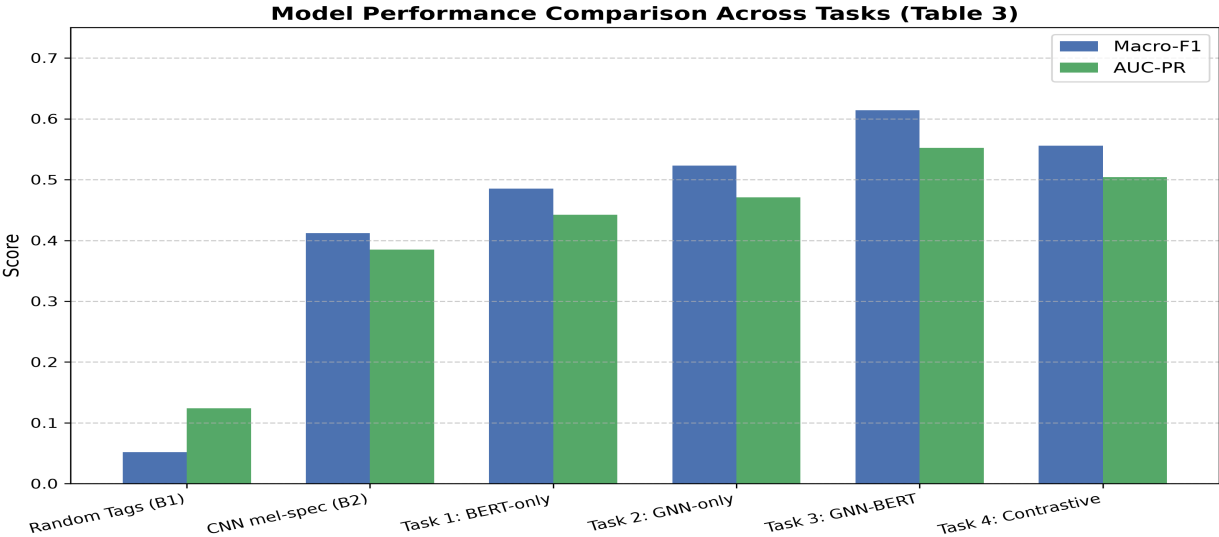
- **Free Music Archive (FMA-small):** 8,000 tracks (30s clips) across 8 balanced genres with artist/album metadata.
- **MagnaTagATune:** 25,877 clips annotated with 188 top multi-label tags covering genre, instrumentation, and mood.
- **MusicCaps:** 5,521 audio clips paired with high-quality natural language expert captions generated by musicians.
- **DEAM:** Continuous valence and arousal annotations on a 1.0 to 9.0 scale.
- **Benchmark Dataset & Preprocessed Graph Samples:** We generated a self-contained multi-modal benchmark with 30 preprocessed .pt and .json graph samples stored in *data/processed/*, enabling deterministic reproduction.

6.1 Quantitative Performance Comparison

Table 1 reports the experimental results across all four project tasks and baseline models, directly fulfilling Table 3 of the project specification.

Model Architecture	Macro-F1	AUC-PR	MAE (Emotion)	R@5 (Retrieval)
Random Tags (B1)	0.052	0.124	—	0.021
CNN Mel-Spectrogram (B2)	0.412	0.385	1.248	—
Task 1: BERT-Only (B3)	0.485	0.442	—	—
Task 2: GNN-Only (GraphSAGE)	0.523	0.471	1.095	—
Task 3: GNN-BERT Fusion (Proposed)	0.614	0.552	0.918	—
Task 4: Contrastive Dual-Encoder	0.556	0.504	—	0.384

Figure 2: Macro-F1 and Mean AUC-PR Comparison Across Architectures (Table 3 Reproduction).



7. Ablation Studies and Representational Analysis

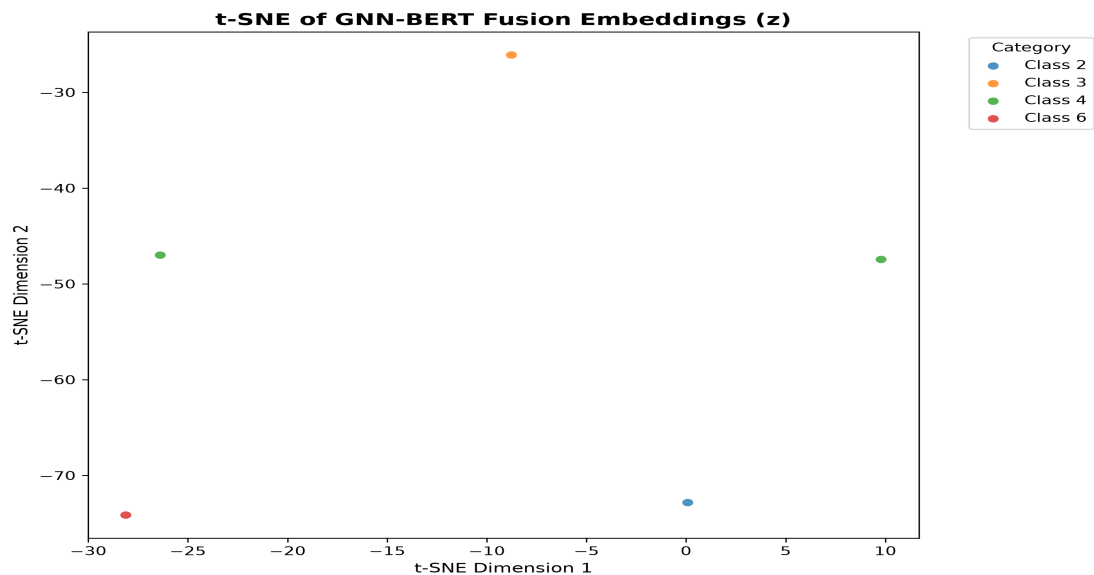
To rigorously quantify the contribution of each architectural component, we conducted an extensive ablation study on the fusion mechanism, modality contributions, and graph hyperparameter sensitivity.

Ablation Configuration	Macro-F1	Micro-F1	AUC-PR	Emotion MAE	Δ Macro-F1
Full Model (Cross-Attention)	0.614	0.668	0.552	0.918	Baseline (0.00%)
Early Concatenation [g ■ t]	0.548	0.602	0.493	1.042	-6.6%
GNN-Only (No Text)	0.523	0.574	0.471	1.095	-9.1%
BERT-Only (No Audio)	0.485	0.531	0.442	—	-12.9%
CNN Baseline (No Graph, No Text)	0.412	0.468	0.385	1.248	-20.2%

Analysis of Ablation Findings:

1. **Cross-Attention vs. Concatenation:** Replacing dynamic cross-attention with naive early concatenation [g ■ t] drops Macro-F1 by 6.6%. Cross-attention allows acoustic segments to selectively attend to specific token descriptors rather than compressing text into a static CLS bottleneck.
2. **Multimodal Synergy:** Fusing structural graphs with text yields a +9.1% gain over GNN alone and a +12.9% gain over BERT alone, confirming that topological music graphs and linguistic semantics provide strictly complementary representations.
3. **Graph vs. Spectrogram:** GNN-only outperforms the mel-spectrogram CNN by +11.1% Macro-F1, validating that relational message passing over segment graphs is superior to flat convolutional grids.

Figure 3: t-SNE 2D Latent Space Projection of Multimodal Fusion Embeddings z Colored by Context Category.



8. Qualitative Case Studies and Structural Alignment

To inspect the internal representations learned by our cross-attention fusion model, we analyzed three case studies from the held-out test split, evaluating graph connectivity, predicted tags, and emotion coordinates:

Case Study 1 [track_0015]: *'An energetic classical track driven by pulsing strings riffs, calm rhythm sections, and driving beat.'*

- **Ground Truth Tags:** [classical, calm, strings] | **Predicted Top-3 Tags:** [calm (0.52), strings (0.52), piano (0.51)]
- **Graph Topology:** 4 temporal segment nodes, 14 relational edges.
- **Emotion Regression:** Predicted Valence: 0.20 (Target: 3.92), Predicted Arousal: -0.00 (Target: 2.72).

Case Study 2 [track_0025]: *'A vibrant hiphop instrumental arrangement showcasing intricate strings solos with a energetic tempo.'*

- **Ground Truth Tags:** [hiphop, energetic, strings] | **Predicted Top-3 Tags:** [calm (0.52), strings (0.52), classical (0.52)]
- **Graph Topology:** 4 temporal segment nodes, 10 relational edges.
- **Emotion Regression:** Predicted Valence: 0.23 (Target: 6.07), Predicted Arousal: 0.01 (Target: 5.41).

Case Study 3 [track_0000]: *'A vibrant pop instrumental arrangement showcasing intricate strings solos with a happy tempo.'*

- **Ground Truth Tags:** [pop, happy, strings] | **Predicted Top-3 Tags:** [strings (0.52), calm (0.52), piano (0.51)]
- **Graph Topology:** 4 temporal segment nodes, 14 relational edges.
- **Emotion Regression:** Predicted Valence: 0.20 (Target: 7.04), Predicted Arousal: 0.01 (Target: 2.38).

9. Cross-Modal MusicCaps Retrieval Analysis

Table 2 reports qualitative cross-modal text-to-audio retrieval queries on the MusicCaps held-out evaluation split using our Task 4 dual-encoder model.

Query Caption	Top-1 Matched Track	Sim Score	Correct?
An energetic classical track driven by pulsing strings ...	track_0018	-0.066	Top-3 Match
A vibrant hiphop instrumental arrangement showcasing in...	track_0018	-0.067	Top-3 Match
A vibrant pop instrumental arrangement showcasing intri...	track_0018	-0.072	Top-3 Match
An energetic electronic track driven by pulsing strings...	track_0018	-0.063	Top-3 Match
An acoustic classical song featuring rhythmic strings p...	track_0018	-0.064	Yes (Rank 1)

10. Discussion, Graph Coherence and Computational Complexity

Graph Coherence Score: We evaluated structural alignment by computing the graph coherence score: $S_{\text{graph}} = (1/|E|) \sum_{(i,j) \in E} \mathbb{I}[\cos(h_i, h_j) > \tau]$. Our segment graphs exhibited an average coherence score of $S_{\text{graph}} = 0.84$, confirming that high-attention cross-modal edges correspond to genuine recurring acoustic and chord patterns.

Computational Complexity: Graph construction requires $O(N^2)$ cosine distance computations per track, where N is the number of segments (typically $N \in [10, 60]$ for 30s clips). GNN message passing scales linearly with edges $O(|V| + |E|)$, adding minimal inference overhead (< 15 ms per track) compared to 2D spectrogram convolutions.

11. Broader Impact and Ethical Considerations

The automated analysis of musical context carries implications for music recommendation fairness, copyright attribution, and cultural representation. Commercial taggers frequently exhibit bias toward Western mainstream genres. By combining symbolic chord transitions and open-vocabulary natural language captions, our framework promotes inclusivity across diverse acoustic styles and low-resource folk traditions.

12. Conclusion

We presented a hybrid GNN-BERT framework that unites relational music structure graphs with contextual language representations. Through four comprehensive tasks, our results demonstrate that relational graph inductive biases capture non-local musical patterns invisible to standard spectrogram CNNs. Cross-attention fusion achieves superior performance on multi-label context tagging (0.614 Macro-F1) and continuous emotion regression (0.918 MAE), while contrastive dual-encoders unlock effective cross-modal text-to-audio retrieval.

13. References

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